

Fig. 7: Program output screenshot of a light ray trapped in the core

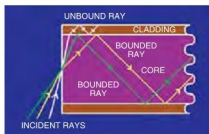


Fig. 8: Program output screenshot of a multimode fibre



Fig. 9: Program output screenshot of a single-mode fibre



Fig. 10: Program output screenshot of closing of the program

screenshots are displayed sequentially on your monitor screen:

1. Optical fibre with its various components (core, cladding, buffer and jacket). These are differentiated with different colours and labeled as well to provide better presentation. Fig. 4 shows program output screenshot of an optical fibre.

2. Block diagram of optical fibre communication link. It includes the main blocks such as source, transmitter, signal regenerator, receiver and destination. The direction of signal flow is also provided in the display. Fig. 5 shows the program output screenshot of the fibre-optic commu-

cation link.

3. Principle of total internal reflection with incidence of light rays at different angles with respect to critical angle and their relative motions from the surface of the two media with different levels of refractive indices. The program shows the flow of rays with delays to enhance presentation. Three cases are taken: when the angle of incidence is less than the critical angle (indicated by yellow rays), when it is equal to the critical angle (indicated by brown rays) and when it is greater than the critical angle (indicated by red rays). Fig. 6 shows the program output screenshot of total internal reflection with rays of incidence at different angles.

4. Optical fibre link with a light ray getting trapped in the core as it is incident at an angle large enough to cause total internal reflection. Proper flow of electrical signal from the transmitter to the receiver is demonstrated here. Fig. 7 shows the program output screenshot of the light ray trapped in the core.

5. Two types of fibres (single-mode and multimode)

Fig. 8 shows the program output screenshot of a multimode fibre. Out of the three incident rays, one is incident out of the acceptance cone and thus turned out as unbounded ray. It gets collapsed in the cladding layer only. The other two rays are within the acceptance cone and therefore undergo total internal reflection.

Fig. 9 shows the program output screenshot of a single-mode fibre. This fibre has a very small diametric core and thus allows only one ray to pass through it.

Fig. 10 shows the program output screenshot of closing of the program. Press any key on the keyboard to exit the program.

Functions used

Initgraph(&gd,&gm,"c:\\turbo3\\bgi"). Initialises the graphics system by loading a graphics driver from the disk and then putting the system into graphics mode. It also resets all the graphics settings like colour, palette and current position to their defaults. 'gd' specifies the graphics driver to be used. 'gm' specifies the initial graphics mode (unless gd=detect). When gd=detect, initgraph sets the graphics mode to the highest resolution available for the detected driver.

Closegraph(). Switches back the screen from graphics mode to text mode. It is called after getch() function so that the screen doesn't clear until the user hits the key.

Void projectwork(). Enables the display of various presentations.

Void closing(). Provides an exit screen before exiting.

Setbkcolor(number). Sets the background colour of choice from 16 colour options 0 through 15. Instead of specifying the number, the respective colour name from the specified set can also be used in block letters.

Setcolor(number). Selects a foreground or drawing colour from the same set of 16 colours.

Settextstyle(font,direction,character size). Sets the values for the text characteristics to be used for displaying texts.

Outtextxy(x,y,"Literal expression"). Displays the string in graphics mode. The (x,y) denotes the starting coordinates of the string on the screen.

Setfillstyle(pattern,color). Fills the colour in the figure. Colours can be filled in a specific pattern.

Floodfill(). Fills the required colour in geometrical figures which can be drawn using the inbuilt functions of shapes (rectangle, circle, etc).

EFY note. The source code of this article is included in this month's EFY-CD and is also available on website www.efymag.com. ●

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